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### SOFTWARE ISOLATION OF VIRTUAL MACHINE RESOURCES

#### Abstract

Methods and systems are provided for isolating resources of a virtual machine (VM) guest from a host operating system (OS). The techniques include receiving, from a guest partition corresponding to an isolated VM guest, an acceptance request that identifies a guest memory page mapped into a guest physical address (GPA) space and a memory page visibility class, e.g., exclusive, shared read-only, or shared read-write. The techniques further include verifying, via a host OS second-level address translation table, that the physical memory page satisfies the specified visibility class by confirming that the host OS has the appropriate access rights. Upon successful verification, the guest memory page is transitioned from an unaccepted state to an accepted state. These techniques enable controlled memory sharing and enhanced isolation between guest VMs and the host OS in virtualized environments.

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 17/837,688, filed Jun. 10, 2022, and entitled “SOFTWARE ISOLATION OF VIRTUAL MACHINE RESOURCES,” and which issued as U.S. Pat. No. \_\_\_\_\_ on \_\_\_\_\_, the entire contents of which are incorporated by reference herein in their entirety.

### BACKGROUND

[0002] Hypervisor-based virtualization technologies allocate portions of a computer system's physical resources (e.g., processor cores and/or time, physical memory regions, storage resources, etc.) into separate partitions, and execute software within each of those partitions. Hypervisor-based virtualization technologies therefore facilitate creation of virtual machine (VM) guests that each executes guest software, such as an operating system (OS) and other software executing therein. While hypervisor-based virtualization technologies can take a variety of forms, many use an architecture comprising a hypervisor that has direct access to hardware and that operates in a separate execution environment than all other software in the system, a host partition that executes a host OS and host virtualization stack, and one or more guest partitions corresponding to VM guests. The host virtualization stack within the host partition manages guest partitions, and thus the hypervisor grants the host partition a greater level of access to the hypervisor, and to hardware resources, than it does to guest partitions.

[0003] Taking HYPER-V from MICROSOFT CORPORATION as one example, the HYPER-V hypervisor is the lowest layer of a HYPER-V stack. The HYPER-V hypervisor provides the basic functionality for dispatching and executing virtual processors for VM guests, but it relies on the HYPER-V host stack for many other aspects of VM guest virtualization. The HYPER-V hypervisor takes ownership of hardware virtualization capabilities (e.g., second-level address translation (SLAT) processor extensions such as Rapid Virtualization Indexing from ADVANCED MICRO DEVICES, or Extended Page Table from INTEL; an input/output (I/O) memory management unit (IOMMU) that connects a direct memory access-capable I/O bus to main memory; processor virtualization controls), and it provides a set of interfaces to allow the HYPER-V host stack to leverage these virtualization capabilities to manage VM guests. The HYPER-V host stack, on the other hand, contains a majority of HYPER-V functionality. The HYPER-V host stack contains components that span kernel and user mode of a host OS executing within the host partition, and that provide general functionality for VM guest virtualization (e.g., memory management, VM guest lifecycle management, device virtualization).

[0004] With current hypervisor-based virtualization architectures, the host OS (and the virtualization stack running therein) assumes it has full access to each VM guest, including all of the VM guest's state. For example, host virtualization stacks expect to be able to read from and write to any portion of a guest partition's memory (e.g., for purposes of device I/O), and expect to be able to read and manipulate a guest partition's processor registers (e.g., for the purposes of device emulation).

[0005] The subject matter claimed herein is not limited to embodiments that solve any disadvantages or that operate only in environments such as those described above. Rather, this background is only provided to illustrate one exemplary technology area where some embodiments described herein may be practiced.

#### BRIEF SUMMARY

[0006] In some aspects, the techniques described herein relate to a method, implemented at a computer system that includes a processor, for isolating resources of a VM guest from a host OS, the method including: receiving an acceptance request from a guest partition corresponding to an isolated VM guest, the acceptance request identifying: a guest memory page that is mapped into a guest physical address (GPA) space of the guest partition, and a memory page visibility class; and setting a page acceptance indication for the guest memory page from an unaccepted state to an accepted state based on a physical memory page mapped to the guest memory page meeting the memory page visibility class.

[0007] In some aspects, the techniques described herein relate to a computer system for isolating resources of a VM guest from a host OS, including: a processor; and a computer storage media that stores computer-executable instructions that are executable by the processor to cause the computer system to at least: receive an acceptance request from a guest partition corresponding to an isolated VM guest, the acceptance request identifying: a guest memory page that is mapped into a GPA space of the guest partition, and a memory page visibility class; and set a page acceptance indication for the guest memory page from an unaccepted state to an accepted state based on a physical memory page mapped to the guest memory page meeting the memory page visibility class.

[0008] In some aspects, the techniques described herein relate to a computer program product including a computer storage media that stores computer-executable instructions that are executable by a processor to cause a computer system to isolate resources of a VM guest from a host OS, the computer-executable instructions including instructions that are executable by the processor to cause the computer system to at least: receive an acceptance request from a guest partition corresponding to an isolated VM guest, the acceptance request identifying: a guest memory page that is mapped into a GPA space of the guest partition, and a memory page visibility class; and set a page acceptance indication for the guest memory page from an unaccepted state to an accepted state based on a physical memory page mapped to the guest memory page meeting the memory page visibility class.

[0009] This summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

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## Description

#### BRIEF DESCRIPTION OF THE DRAWINGS

[0010] In order to describe the manner in which the advantages and features of the systems and methods described herein can be obtained, a more particular description of the embodiments briefly described above will be rendered by reference to specific embodiments thereof which are illustrated in the appended drawings. Understanding that these drawings depict only typical embodiments of the systems and methods described herein, and are not therefore to be considered to be limiting of their scope, certain systems and methods will be described and explained with additional specificity and detail through the use of the accompanying drawings in which:

[0011] FIG. 1 illustrates an example computer architecture that facilitates software isolation of VM resources;

[0012] FIG. 2 illustrates an example of an isolated VM (IVM) security component; [0013] FIG. 3 illustrates an example of phases of an IVM guest lifecycle; and [0014] FIG. 4 illustrates a flow chart of an example method for isolating resources of a VM guest from a host OS.

## DETAILED DESCRIPTION

[0015] In cloud computing architectures, customers' VM guests and the data they operate on are hosted by cloud providers within virtualized environments. As cloud computing becomes commonplace, providing strong assurances around the confidentiality and integrity of hosted VM guests, and their data, is of increased importance to cloud providers and their customers. For example, being able to provide strong assurances around the security and privacy of hosted VM guests enables security-focused customers to move their workloads to the cloud. In addition, providing stronger isolation and protection of VM guests can also be beneficial to on-premises customers that manage their own data centers, because stronger isolation and protection of VM guests protects against malicious infiltrators into an on-premises virtualization fabric.

[0016] At least some embodiments described herein are directed to a software-based architecture, referred to herein as an IVM architecture, to provide IVM guests. The IVM architecture enables highly secure and confidential computing by fully isolating VM guest state (e.g. registers, memory) from a host operating system (OS) executing within a host partition, as well as from an entity that manages a computing system on which the IVM guests are hosted. To achieve the foregoing, the IVM architecture described herein introduces a new security boundary between the hypervisor and the host virtualization stack. In embodiments, this new security boundary is enforced by the hypervisor, by restricting which VM guest resources can be accessed by the host OS (and, in turn, the host virtualization stack) to ensure the integrity and confidentiality of an IVM guest.

[0017] In addition to these security benefits, in embodiments, the IVM architecture described herein provides a benefit of supporting existing guest OSs (e.g. without requiring changes to the guest OS kernel and boot loader). For example, in embodiments, existing guest OSs can be supported via enlightened OS drivers. Additionally, because it is software-based, the IVM architecture described herein is operable on existing computing hardware. Thus, for example, cloud providers can leverage the security benefits described herein without needing to wait for the development, or purchase, of new hardware systems that are designed with hardware-based VM guest isolation.

[0018] FIG. 1 illustrates an example computer architecture **100** (e.g., including the IVM architecture) that facilitates software isolation of VM guest resources. As shown, computer architecture **100** includes hardware **101** (e.g., a computer system) comprising a processor **102** (e.g., a single processor, or a plurality of processors), memory **103** (e.g., system or main memory), storage media **104** (e.g., a single computer-readable storage medium, or a plurality of computer-readable storage media), an IOMMU **105**, and firmware **106** (e.g., stored on storage media **104**, and/or stored on a dedicated memory device such as a read-only memory). Although not shown, hardware **101** may also include other hardware devices, such as a trusted platform module (TPM) for facilitating measured boot features, a network interface (e.g., one or more network interface cards) for interconnecting (via a network) to one or more other computer systems, video display interface(s), user input interface(s), and the like.

[0019] As shown, in computer architecture **100**, a hypervisor **107** executes directly on hardware **101**. In general, the hypervisor **107** partitions hardware resources (e.g., processor **102**, memory **103**, I/O resources) among a host partition **111** that executes a host OS **116**, as well as a guest partition **112** (or, as shown, a plurality of guest partitions) that executes a guest OS **118**. In the description herein, the term “VM guest” is used to refer to the “guest partition,” and the term “IVM guest” is used to indicate when a VM guest is an isolated VM guest operating in an isolated guest partition under the IVM architecture described herein. The hypervisor **107** also enables regulated communications between partitions via a bus **113** (e.g., a VM BUS). As shown, the host OS **116**

includes a virtualization stack **117** which manages VM guest virtualization (e.g., memory management, VM guest lifecycle management, device virtualization) via one or more application program interface (API) calls to the hypervisor **107**.

[0020] Computer architecture **100** includes a security component **108**, which provides functionality for transforming any VM guest, such as guest partition **112**, into an IVM guest by isolating resources (e.g., registers within processor **102**, portions of memory **103**, I/O resources) allocated to the VM guest from the host OS **116** (and, in turn, from the virtualization stack **117**). In embodiments, the security component **108** has privileged access to VM guest state. In computer architecture **100**, the security component **108** is shown as operating within the hypervisor **107**; thus, in some embodiments, functionality of the security component **108** is partially or fully implemented within the hypervisor **107**.

[0021] In computer architecture **100**, the host partition **111** is shown as comprising a higher trust zone **114** and a lower trust zone **115**. Here, the host OS **116** operates within the lower trust zone **115** and the security component **108** operates within the higher trust zone **114**; thus, in some embodiments, functionality of the security component **108** is partially or fully implemented within the higher trust zone **114**. In embodiments, the higher trust zone **114** is isolated from the lower trust zone **115** based at least on mappings within SLAT tables **109** that map system physical addresses (SPAs) to guest physical address (GPAs), and within IOMMU translation tables (IOMMU tables **110**). As a result, the higher trust zone **114** (and the security component **108**) is isolated from the lower trust zone **115** (and the host OS **116** and software executing thereon such as the virtualization stack **117**). In one example, the hypervisor **107** is the HYPER-V hypervisor and supports virtualization-based security (VBS) to sub-partition partitions into virtual trust levels (VTLs), the higher trust zone **114** operates under VBS in a higher privileged VTL 0, and the lower trust zone **115** operates under VBS in a lower privileged VTL 1.

[0022] In embodiments, using the security component **108**, the IVM architecture introduces a new security boundary between the hypervisor **107** and the host OS **116**. This new security boundary is illustrated using a heavy line surrounding the lower trust zone **115**, and a heavy line surrounding the IOMMU **105**. In embodiments, due to this new security boundary, the only components of computer architecture **100** that are within a virtualization trusted computing base (TCB), upon which an IVM guest depends, are trusted firmware components within firmware **106**, the hypervisor **107**, and components within the higher trust zone **114** of the host partition **111** (if present). In an implementation using the HYPER-V stack, and using WINDOWS as the host OS **116**, introduction of this new security boundary reduces the number of lines of code within the virtualization TCB by over 95%.—significantly reducing the potential attack surface. In addition to reducing a size of the virtualization TCB, the IVM architecture described herein also provides a well-defined and scoped security boundary between the virtualization TCB and the rest of the system. This security boundary is defensible, and greatly simplifies validation of a virtualization host's TCB.

[0023] Consistent with this new security boundary, in computer architecture **100** the hypervisor **107** and the higher trust zone **114** (including components operating within, such as the security component **108**) are illustrated as being trusted and within the virtualization TCB, while the lower trust zone **115** (including components operating within, such as the host OS **116** and the virtualization stack **117**) are illustrated as being untrusted and outside of the virtualization TCB. Notably, in embodiments in which the host partition **111** lacks a separation into a higher trust zone **114** and a lower trust zone **115** (and, thus, the security component **108** is fully implemented within the hypervisor **107**), then the entirety of the host partition **111** would be within the virtualization TCB. Either way, with this new security boundary, the host OS **116** is considered to be untrusted and is outside of the virtualization host TCB. This provides a well-defined and defensible security boundary that strongly isolates IVM guests from the host OS **116**.

[0024] Additionally, DMA devices behind the IOMMU **105** are illustrated as being outside of the

virtualization TCB, and the firmware **106** is illustrated as being both partially within and partially outside of the virtualization TCB. In embodiments, the amount of the firmware **106** that is within the virtualization TCB varies based on whether the computer architecture **100** supports Dynamic Root of Trust for Measurement (D-RTM). On systems that leverage D-RTM, the firmware components within the virtualization TCB can be limited to microcode of the processor **102** and the firmware used for D-RTM launch. This is because D-RTM provides the ability to eliminate a Basic Input Output System (BIOS) and runtime firmware components—including, e.g., System Management Mode (SMM) and Unified Extensible Firmware Interface (UEFI) runtime services—from the virtualization TCB. On systems that don't support D-RTM, the BIOS and runtime firmware components may be within the virtualization TCB.

[0025] In embodiments, with the IVM architecture described herein, the hypervisor **107** continues to allow the host OS **116** to control the assignment of processor, memory, and I/O resources to a VM guest as would be conventional. This means that the host OS **116** is still responsible for governing resource usage of IVM guests, and that the host OS **116** can use existing allocation policies for controlling how much compute resources an IVM guest receives. This also means that the host OS **116** can potentially deny access to resources by an IVM guest, and thus prevent an IVM guest from performing well, or performing at all. Thus, under the IVM architecture, the host OS **116** still maintains control of allocation of resources among the hardware **101**. However, with the new security boundary introduced by the IVM architecture, the hypervisor **107** restricts the host OS **116** from reading or modifying the contents of an IVM guest's state. This includes restricting the host OS **116** from accessing the contents of an IVM guest's virtual processor state, as well as restricting the host OS **116** from accessing an IVM guest's memory contents.

[0026] FIG. 2 illustrates an example **200** of an IVM security component, such as security component **108** of FIG. 1. Each internal component of the security component **108** depicted in FIG. 2 represents various functionalities that the security component **108** might implement in accordance with various embodiments described herein. It will be appreciated, however, that the depicted components—including their identity and arrangement—are presented merely as an aid in describing example embodiments of the security component **108**. Notably, the internal component of the security component **108** could be distributed in a variety of manners between the higher trust zone **114** and/or the hypervisor **107**.

[0027] In example **200**, the security component **108** includes an attestation component **201**. In embodiments, the attestation component **201** provides attestation and key management functionality that enables a party operating the computer architecture **100** (a hoster) to be fully removed from the virtualization host's TCB. Because the virtualization stack **117** is not trusted, the virtualization stack **117** cannot be relied upon to ensure that an IVM guest is not tampered with prior to the IVM guest being launched. Furthermore, the guest OS **118** running inside of an IVM guest cannot simply query the hypervisor **107** to determine if it was launched securely, as it can never guarantee that it's running on a secure host. Thus, in order for an IVM guest to launch securely, a remote server is responsible for attesting to the launch of the IVM guest and releasing the IVM guest's secrets (e.g. encryption keys) to the software running inside of the IVM guest. Thus, using the attestation component **201**, a tenant operating an IVM guest can securely verify the authenticity and security of a virtualization host on which the tenant's IVM guest operates, and the tenant can release keys to the virtualization host and be assured the party operating the computer architecture **100** is unable to access the keys or read/modify the IVM guest's state without detection.

[0028] In example **200**, the security component **108** also includes a processor state isolation component **202**. Currently, many hypervisors are responsible for managing a VM guest's processor state, and allow the host OS the ability to read and modify a VM guest's processor state for the purposes of emulating devices. In embodiments, the processor state isolation component **202** fully isolates an IVM guest's processor state from the host OS **116**. In particular, the processor state

isolation component **202** prevents the host OS **116** from reading from or writing to an IVM guest's registers-including the IVM guest's instruction and stack pointers.

[0029] In embodiments, the processor state isolation component **202** also prevents the host OS **116** from setting intercepts for events that could result in leaking sensitive information to the host OS **116**. In embodiments, the processor state isolation component **202** also limits the kinds of interrupts and exceptions the host OS **116** can generate for an IVM guest. In one example, the processor state isolation component **202** allows for synthetic interrupts to be generated by the host OS **116** for the purposes of para-virtualized communication via the bus **113**. However, the processor state isolation component **202** does not allow for emulated interrupts or exceptions to be generated by the host OS **116**, because allowing the host OS **116** the ability to generate emulated interrupts or exceptions would present an attack surface, as an IVM guest would be susceptible to unexpected interrupts and exceptions at arbitrary points during its execution.

[0030] In example **200**, the security component **108** also includes a memory isolation component **203**. In embodiments, the memory isolation component **203** isolates the memory of an IVM guest from components within outside of the security boundary illustrated in FIG. **1**. This includes isolating the memory of an IVM guest from the host OS **116** and from DMA devices behind the IOMMU **105**. Additionally, in embodiments, the memory isolation component **203** protects the integrity of each IVM guest's GPA space (e.g., the address space that the IVM guest sees as its own physical address space).

[0031] In computer architecture **100**, the host OS **116** is responsible for memory management (e.g., via the virtualization stack **117**). Thus, the host OS **116** is responsible for allocating physical memory for IVM guests, and assigning physical memory to VM guests (e.g., guest partition **112**) corresponding to IVM guests. The host OS **116** also controls the GPA space for an IVM guest, and decides where to map pages in an IVM guest's GPA space. However, to ensure isolation and integrity of an IVM guest's memory, the memory isolation component **203** enforces a set of rules on the host OS **116** when managing an IVM guest's memory. In embodiments, these rules are centered around memory access controls (MACs) and GPA integrity controls. Thus, the memory isolation component **203** is illustrated as including a MAC component **204** and a GPA integrity control component **205**. In embodiments, the MAC component **204** provides facilities for controlling which portions of memory the host OS **116** can access for an IVM guest, while the GPA integrity control component **205** provides facilities that enable an IVM guest to be assured of its GPA space integrity at all times.

[0032] Referring to the MAC component **204**, for memory pages that have been assigned to an IVM guest, the IVM architecture introduces the concept of memory page visibility classes. In embodiments, the MAC component **204** records host OS visibility for a memory page as one or more per-page host visibility attributes within the SLAT tables **109**. For memory pages that are mapped into an IVM guest's GPA space, these per-page host visibility attribute(s) define the amount of access that the host OS **116** is allowed to each memory page. In embodiments, these memory page visibility classes include exclusive memory, shared read-only memory, and shared read/write memory. In embodiments, these memory page visibility classes define various attributes around the accessibility and management of a memory page, including the host visibility attribute(s) of a memory page.

[0033] Referring to exclusive memory, in embodiments, exclusive memory is physical memory (e.g., within memory **103**) that has been allocated by the host OS **116** for exclusive use by an IVM guest. Thus, conceptually, an IVM guest can be thought of as “owning” any physical memory pages that have been allocated to it as exclusive memory. In embodiments, an IVM guest gets full access privileges (e.g., read, write, and execute) to physical memory pages that have been allocated to it as exclusive memory. In embodiments, the host OS **116** allocates physical memory to be used for an IVM guest's memory, and then it makes one or more API calls to the hypervisor **107** to assign physical memory for use as an IVM guest's memory. When the host OS **116** assigns a

memory page as exclusive to an IVM guest, the MAC component **204** removes the host OS **116**'s access to that memory page. This ensures the contents of an IVM guest's exclusive memory pages are not accessible by the host OS **116**. Thus, the host OS visibility for IVM guest's exclusive memory pages is no access. In embodiments, the MAC component **204** enforces host OS visibility by controlling SLAT tables **109** for the host OS **116** (e.g., host second-level page tables) and the IOMMU tables **110** for DMA devices. Thus, when the host OS **116** allocates a physical memory page and assigns it to an IVM guest for exclusive use by the IVM guest, the MAC component **204** updates SLAT tables **109** for the host OS **116** to prevent access to the memory page by the host OS **116**, and updates IOMMU tables **110** to prevent access to the memory page by DMA devices and their downstream consumers.

[0034] In embodiments, the host OS **116** cannot regain ownership/control of an IVM guest's exclusive memory page until the memory page is successfully unmapped from the IVM guest's GPA space. In embodiments, as part of unmapping an exclusive memory page from the IVM guest's GPA space, the MAC component **204** wipes/clears the contents of the memory page (e.g., by zeroing the memory page, by writing a pattern of bits to the memory page, by writing random bits to the memory page, by deleting an encryption key associated with the memory page).

[0035] In embodiments, the MAC component **204** enforces a set of rules to ensure the privacy and integrity of exclusive memory. In embodiments, these rules require that an IVM guest's exclusive memory page is only mapped/owned by a single VM guest at a time, that an IVM guest's exclusive memory page is only mapped to a single GPA of the IVM guest, that the contents of an IVM guest's exclusive memory page are wiped before the memory page is released back to the host OS **116**, and that an IVM guest's exclusive memory pages are wiped on system reset (e.g., reset of the IVM guest).

[0036] As noted, exclusive memory pages are not visible to the host OS **116** by default. However, there are cases where the guest OS **118** inside of an IVM guest may need to give the host OS **116** access to some of its exclusive memory pages (e.g., for the purposes of I/O). To facilitate this, in embodiments the MAC component **204** allows an IVM guest to change host visibility on a per-page basis for exclusive memory pages. In an example, the MAC component **204** exposes an API to the IVM guest that allows the IVM guest to call the hypervisor **107** to control the host visibility for its exclusive memory pages (e.g., via call(s) from a guest enlightenment **119**, such as a driver operating in the guest OS **118**). In embodiments, when an IVM guest grants the host visibility to an exclusive memory page, or removes host visibility from a memory page, the MAC component **204** updates SLAT tables **109** for the host OS **116** and IOMMU tables **110** to grant or deny the host OS **116** access to the memory page.

[0037] Referring to shared read-only memory, shared read-only memory is defined as host OS **116** memory that is read-only both to the host, as well as to one or more VM guests. This makes shared read-only memory appropriate for scenarios like direct map, in which the host OS **116** wishes to direct map image pages into one or more VM guests. In embodiments, in order to map a shared read-only page into a VM guest, the host OS **116** first designates the memory page as read-only by invoking a call to inform the hypervisor **107**. The host OS **116** then maps the memory page into a VM guest as a shared read-only page. In embodiments, while mapping a shared read-only memory page into a VM guest, the host OS **116** continues to have read access the memory page. Thus, in embodiments, a shared read-only memory page is always host visible for read access, and a VM guest cannot change the host visibility attribute for a shared, read-only memory page. In embodiments, a shared read-only memory page can be mapped into multiple VM guests, and it can be mapped at multiple different GPAs within the same VM guest.

[0038] Referring to shared read-write memory, shared read-write memory is defined as host OS **116** memory that is read-write both to the host OS **116** as well as to VM guests. Thus, a shared read-write page is always host visible for read and write access, and a VM guest cannot change the host visibility attribute for a shared read-write page. As will be appreciated, a VM guest can make



no assumptions about the contents of shared read-write memory, as the host OS **116** can changing the contents at any time. In embodiments, shared read-write memory is used by the host OS **116** for setting up memory regions for sharing data between a VM guest and the host OS **116**.

[0039] Referring to the GPA integrity control component **205**, the GPA integrity control component **205** enables IVM guests to have some guarantees around the integrity and behavior of their GPA space. To ensure the integrity of an IVM guest's memory, the IVM architecture introduces the concept of “acceptance” for memory pages that are mapped into an IVM guest's GPA space. With this acceptance model, an IVM guest accepts a memory page prior to accessing it. In embodiments, if an IVM guest attempts to access a memory page that it has not previously accepted, the hypervisor **107** generates a fault into the guest (e.g., on X64 platforms, the hypervisor **107** may generate a #VE exception).

[0040] In embodiments, an IVM guest accepts a memory page by making one or more API calls (e.g., using the guest enlightenment **119**) that are handled by the GPA integrity control component **205**, specifying the GPA to accept and an expected attribute or attributes of the GPA (e.g., indicating a class of memory it expects for the memory page at the GPA). The GPA integrity control component **205** then determines if the physical memory page mapped by the host OS **116** to the specified GPA meets the criteria of the indicated class of memory, and marks the memory page as accepted (when the GPA meets the criteria of the indicated class of memory) or unaccepted (when the GPA does not meet the criteria of the indicated class of memory). In embodiments, the GPA integrity control component **205** marks a memory page as accepted or unaccepted by setting or clearing a flag or bit in SLAT tables **109** for the IVM guest. In one embodiment, a value of one indicates acceptance, and a value of zero indicates a lack of acceptance. Thus, if the IVM guest is expecting one class of memory to be mapped at a particular GPA, but the host OS **116** has actually mapped a different class of memory, the accept operation will fail resulting in the memory page being marked as unaccepted. In embodiments, the hypervisor **107** prohibits the host OS **116** from marking a memory page as accepted; thus, only IVM guests, via the GPA integrity control component **205**, are capable of accepting memory pages on their own behalf.

[0041] In embodiments, the GPA integrity control component **205** also protects against the host OS **116** maliciously remapping an IVM guest's pages while the IVM guest is using the memory pages, which can cause undetected corruption or unexpected behavior. For example, whenever the host OS **116** changes which physical memory page maps to a GPA for an IVM guest, the GPA integrity control component **205** sets the accepted state of the GPA memory page to unaccepted. Thus, if the host OS **116** attempts to remap a GPA from one SPA to a different SPA, the GPA integrity control component **205** clears the accepted state of the memory page, and any future attempt by the IVM guest to access the GPA will result in a fault.

[0042] As will be appreciated in view of the description herein, this memory acceptance model may not prevent the host OS **116** from maliciously modifying an IVM guest's GPA space; however, this memory acceptance model does ensure that an IVM guest is made aware of any unauthorized changes to its GPA space. For example, an IVM guest can accept all of its addressable memory pages during boot, and begin using its memory. Then, if the host OS **116** changes an SPA to GPA mapping for the IVM guest, the IVM guest will be notified as the next access to the GPA will generate a fault. In embodiments, the acceptance model allows an IVM guest to coordinate with the host OS **116** when it wishes to allow for changes to occur in its GPA space for scenarios like adding and removing memory.

[0043] In example **200**, the security component **108** also includes a memory paging component **206**. In embodiments, memory paging is a special case where IVM architecture allows the host OS **116** to make changes to a running IVM guest's GPA space without coordination with the guest OS **118**. To facilitate memory paging, the memory paging component **206** enforces a defined paging flow in which the host OS **116** cooperates. In particular, the memory paging component **206** records integrity information (e.g., a hash, a checksum) of a memory page's contents during page-

out, and then uses this integrity information to validate the contents of a memory page on page-in, which protects against the host OS **116** maliciously modifying the page contents. In some embodiments, such as when paging memory pages that are not host visible (e.g. exclusive memory pages that aren't host visible), the memory paging component **206** also encrypts memory page contents of during page-out, and decrypts the memory page contents of the page during page-in (e.g., using an encryption key specific to the IVM guest).

[0044] In some environments, when a VM guest boots, the host OS injects boot firmware (e.g., a UEFI BIOS) into the VM guest's memory to “boot” the OS in the VM guest. Thus, to a guest OS inside of a VM guest, the boot process looks substantially the same as it does on a physical machine, and the same OS boot loaders can be used in VM guests as on a physical machine. However, the IVM architecture described herein moves the host OS **116** out of an IVM guest's TCB, so the host OS **116** can no longer inject arbitrary firmware into the IVM guest, as this would be insecure. In addition, the host OS **116** cannot be relied upon to serve as the root of trust for launch of the IVM guest for attestation purposes. A design principle of IVM guests is that the tenant has full control over all of the code running in the IVM guest. Thus, in the IVM architecture described herein, the tenant has control over the firmware running inside of their IVM guests. Furthermore, the IVM architecture provides the tenant a way of validating that the firmware loaded in their IVM guests is the proper firmware through measurement and attestation of the firmware.

[0045] In example **200**, the security component **108** also includes a firmware injection component **207**. In embodiments, the firmware injection component **207** facilitates firmware injection within the IVM architecture, and allows for different models for injecting boot firmware into an IVM guest. In a first model, a tenant owning an IVM guest relies on the host OS **116** to provide a known good boot firmware image. In order to make this first model secure, the tenant needs to be confident that the boot firmware provided by the host OS **116** a known good image, and that it doesn't contain malicious/compromised code. In embodiments, this is accomplished by providing open-source boot firmware images built from a public open-source repository. This allows for tenants that wish to rely on provided boot firmware to audit the firmware's contents and know exactly what's contained inside of the boot firmware image. In other embodiments, a tenant owning an IVM guest provides their own boot firmware image to be used for the IVM guest. In embodiments, in order to distribute boot firmware, a defined boot firmware image format includes a binary firmware blob to load into the IVM guest's memory prior to IVM guest start-up, and metadata that specifies, e.g., where the boot firmware image should be loaded in the IVM guest's GPA space, and an offset into the boot firmware image for where execution should start.

[0046] FIG. **3** illustrates an example **300** of phases of an IVM guest lifecycle. As shown in example **300**, an IVM guest lifecycle includes a phase **301** of creating an isolated guest partition (e.g., guest partition **112**), in which the virtualization stack **117** invokes APIs of the hypervisor **107** to create guest partition **112** (an IVM guest). During this initialization phase of the IVM guest, the virtualization stack **117** goes through the conventional steps for initializing a partition with the hypervisor **107**, including allocating and assigning physical memory to the IVM guest. As will be appreciated in view of the foregoing description of the MAC component **204** and the GPA integrity control component **205**, all memory pages assigned to the IVM guest will not have yet been accepted by the IVM guest, and thus these memory pages will not yet be accessible by the IVM guest.

[0047] The IVM guest lifecycle also includes a phase **302** of obtaining IVM guest boot firmware (e.g., boot firmware **120**). As shown, there is no ordering required between phase **301** and phase **302**. At phase **302**, the virtualization stack **117** is provided with an initial boot firmware image for the IVM guest to load into memory. This initial boot firmware image includes a bootloader, which is where the IVM guest will begin executing. This initial boot firmware image is either an image specified by the IVM guest's owner, or a standard image provided by the host OS **116**. This allows a user to specify their own boot firmware image or use a standard boot firmware image provided by

a virtualization product or host.

[0048] The IVM guest lifecycle also includes a phase **303** of associating an IVM security component with the isolated guest partition. At phase **303**, the virtualization stack **117** launches an instance of the security component **108** for the IVM guest, and associates this instance with the IVM guest.

[0049] The IVM guest lifecycle also includes a phase **304** of providing the IVM guest boot firmware to the IVM security component. At phase **304**, virtualization stack **117** provides the boot firmware image to the security component **108**, and designates where the boot firmware image should be loaded in the IVM guest's GPA space.

[0050] The IVM guest lifecycle also includes a phase **305** of copying the IVM guest boot firmware to isolated guest memory. At phase **305**, the firmware injection component **207** copies the boot firmware image into the appropriate place in the IVM guest's GPA space (e.g., boot firmware **120**). In connection with this, the firmware injection component **207** (or the GPA integrity control component **205**) changes the acceptance state of the GPA memory pages containing the boot firmware image to be "accepted," thus allowing the IVM guest to access these boot firmware image memory pages.

[0051] Once the boot firmware has fully been established, the virtualization stack **117** indicates to the hypervisor **107** that the IVM guest is now ready to begin execution. At this point, the virtualization stack **117** is no longer be able to manipulate the configuration of the IVM guest or access the IVM guest's exclusive memory.

[0052] The IVM guest lifecycle also includes a phase **306** of measuring the IVM guest configuration. At phase **306**, the attestation component **201** measures the configuration of the IVM guest, including the IVM guest's initial boot firmware image.

[0053] The IVM guest lifecycle also includes a phase **307** of beginning IVM guest execution. At phase **307** the security component **108** begins execution of the IVM guest at the start of the IVM guest's initial bootloader. Thus, as this phase, the IVM guest is running. The only memory pages that the IVM guest can access are the memory pages that were initialized as part of injecting the initial boot firmware image.

[0054] The IVM guest lifecycle also includes a phase **308** of managing an IVM guest memory page acceptance model. At phase **308**, the IVM guest accepts additional memory pages using API call(s) handed by the GPA integrity control component **205**. In some embodiments, this involves the initial bootloader querying the host OS **116** for memory configuration information, and then accepting pages based on the IVM guest's memory configuration reported by the host OS **116**. As noted in connection with the description of the acceptance model, if there is a disparity in what the host OS **116** reports and what the IVM guest is able to accept, this will be detected during the accept, as the acceptance will fail.

[0055] Although not expressly illustrated, when an IVM guest is stopped, its memory will be reclaimed from the hypervisor **107** by the virtualization stack **117**. As part of the reclamation, the MAC component **204** will wipe the contents of all of the IVM guest's exclusive memory pages prior to granting the host OS **116** access to the memory contents.

[0056] Operation of computer architecture **100** is now described in connection with FIG. **4**, which illustrates a flow chart of an example method **400** for isolating resources of a VM guest (e.g., guest partition **112**) from a host OS (e.g., host OS **116** within host partition **111**). In embodiments, instructions for implementing method **400** are encoded as computer-executable instructions (e.g., security component **108**) stored on a computer storage media (e.g., storage media **104**) that are executable by a processor (e.g., processor **102**) to cause a computer system (e.g., hardware **101**) to perform method **400**.

[0057] The following discussion now refers to a number of methods and method acts. Although the method acts may be discussed in certain orders, or may be illustrated in a flow chart as occurring in a particular order, no particular ordering is required unless specifically stated, or required because

an act is dependent on another act being completed prior to the act being performed.

[0058] Referring to FIG. 4, method **400** comprises an act **401** of receiving a guest memory page acceptance request. In some embodiments, act **401** comprises receiving an acceptance request from a guest partition corresponding to an isolated VM guest (IVM guest). In embodiments, the acceptance request identifies (1) a guest memory page that is mapped into a GPA space of the guest partition, and (2) a memory page visibility class. In an example, the GPA integrity control component **205** receives an API call from the guest partition **112** (e.g., originating from the guest enlightenment **119**) requesting to accept a memory page (e.g., in reference to a GPA of the memory page) with an expected memory page visibility class (e.g., exclusive visibility, shard read-only visibility, shared read-write visibility).

[0059] Method **400** also comprises an act **402** of determining whether a mapped physical memory page meets a requested memory page visibility class. In some embodiments, act **402** comprises determining whether a physical memory page that is mapped to the guest memory page meets the memory page visibility class. In an example, the GPA integrity control component **205** determines what physical memory page maps to the guest memory page indicate by the IVM guest, what memory page visibility class the IVM guest expects, and whether the physical memory page meets that memory page visibility class.

[0060] As shown, act **402** includes an act **403** when the memory page visibility class is exclusive visibility (e.g., when the memory page visibility class is the exclusive visibility class), or an act **406** when the requested memory page visibility class is shared visibility (e.g., when the memory page visibility class is the shared read-only visibility class, or when the memory page visibility class is the shared read-write visibility class).

[0061] When the requested memory page visibility class is exclusive visibility (act **403**), method **400** includes an act **404** of verifying guest exclusivity (via SLAT). In embodiments, act **404** comprises verifying, via one or more guest SLATs, that the physical memory page is exclusively mapped to the guest memory page. For example, as was described, the MAC component **204** enforces a set of rules to ensure the privacy and integrity of exclusive memory, including requiring that an IVM guest's exclusive memory page is only mapped/owned by a single VM guest at a time (e.g., within the SLAT tables **109**), and requiring that an IVM guest's exclusive memory page is only mapped to a single GPA of the IVM guest (e.g., within the SLAT tables **109**). Thus, in some embodiments, the GPA integrity control component **205** verifies that each of these conditions are true for the physical memory page that is mapped to the requested guest memory page.

[0062] When the requested memory page visibility class is exclusive visibility (act **403**), method **400** also includes act **405** of verifying no host access (via SLAT). In embodiments, act **405** comprises verifying, via a host OS SLAT, that the host OS is denied access to the physical memory page. For example, as was described, the MAC component **204** enforces host OS visibility by controlling SLAT tables **109** for the host OS **116**. Thus, when the host OS **116** allocates a physical memory page and assigns it to an IVM guest for exclusive use by the IVM guest, the MAC component **204** updates SLAT tables **109** for the host OS **116** to prevent access to the memory page by the host OS **116**.

[0063] Although not shown in FIG. 4, as described, the MAC component **204** can also enforce host OS visibility to a memory page by controlling the IOMMU tables **110** for DMA devices. Thus, some embodiments of act **403** further comprise verifying, via host OS IOMMU table, that a DMA device is denied access to the physical memory page.

[0064] As was mentioned in connection with the description of exclusive visibility, there are cases where the guest OS **118** inside of an IVM guest may need to give the host OS **116** access to some of its exclusive memory pages (e.g., for the purposes of I/O), and to facilitate this the MAC component **204** allows an IVM guest to change host visibility on a per-page basis for exclusive memory pages. Thus, some embodiments of method **400** further comprise receiving a visibility change request from the guest partition, the visibility change request including an indication of the

guest memory page; and updating the host OS SLAT to grant physical memory page access to the host OS.

[0065] In embodiments where the requested memory page visibility class is shared visibility (act **403**) such as shared read-only visibility or shared read-write visibility, method **400** includes an act **407** of verifying host shared access (via SLAT). In some embodiments, the memory page visibility class is the shared read-only visibility class, and act **407** comprises verifying that the physical memory page meets the memory page visibility class, including verifying, via a host OS SLAT, that the host OS is granted read-only access to the physical memory page. In other embodiments, the memory page visibility class is the shared read-write visibility class, and act **407** comprises verifying that the physical memory page meets the memory page visibility class, including verifying, via a host OS SLAT, that the host OS is granted read-write access to the physical memory page.

[0066] Depending on the outcome of act **402**, method **400** also comprises either act **408** of accepting the memory page (when act **402** determines that the mapped physical memory page meets the requested memory page visibility class), or an act **409** of rejecting the memory page (when act **402** determines that the mapped physical memory page does not meet the requested memory page visibility class). In some embodiments, act **408** comprises setting a page acceptance indication for the guest memory page from an unaccepted state to an accepted state based on the physical memory page mapped to the guest memory page meeting the memory page visibility class. In some embodiments, act **409** comprises, setting a page acceptance indication for the guest memory page to an unaccepted state based on the physical memory page mapped to the guest memory page not meeting the memory page visibility class.

[0067] As a result of method **400**, an IVM guest accepts the physical memory page mapped to an identified guest memory page only when that physical memory page meets an expected memory page visibility class. In this way, the IVM guest is aware of the security implications of using the identified guest memory page, such as by knowing whether the contents of the guest memory page are exclusive to the IVM guest, or whether the contents of the guest memory page may be read from and/or written to by the host OS **116** and/or by another VM guest. This provides the IVM guest the ability to isolate its data from the host OS **116**.

[0068] As mentioned, the GPA integrity control component **205** protects against the host OS **116** maliciously remapping an IVM guest's memory pages while the IVM guest is using the memory pages. This protection comes by setting the accepted state of the memory page to unaccepted when the host OS **116** changes the physical memory page to which a GPA maps for an IVM guest. Thus, in some embodiments, method **400** further comprises detecting that the guest memory page has been mapped to a different physical memory page; and setting the page acceptance indication for the guest memory page to the unaccepted state.

[0069] As mentioned, in embodiments where an IVM guest attempts to access a memory page that it has not previously accepted, the hypervisor **107** generates a fault into the guest (e.g., on X64 platforms, this may be a #VE exception). Thus, in some embodiments, method **400** also comprises, prior to receiving the acceptance request, detecting an access by the guest partition to an address covered by the guest memory page; and generating a page fault based on the page acceptance indication for the guest memory page being the unaccepted state.

[0070] As discussed, the memory paging component **206** enforces a defined paging flow, in which the memory paging component **206** records integrity information (e.g., a hash, a checksum) of a memory page's contents during page-out by the host OS **116**, and then uses this integrity information to validate the contents of pages on page-in by the host OS **116**, thereby protecting against the host OS **116** maliciously modifying the page contents. As such, in some embodiments, method **400** also comprises, based on receiving a page-out request from the host OS that identifies the physical memory page, recording integrity information for contents of the physical memory page; and based on receiving a page-in request from the host OS that includes an indication of the

physical memory page, using the integrity information to verify the contents of the physical memory page.

[0071] Additionally, as discussed, the memory paging component **206** may also encrypt memory page contents of during page-out, and decrypt the memory page contents of the page during page-in. Thus, in some embodiments, method **400** also comprises, based on receiving the page-out request from the host OS, encrypting the contents of the physical memory page; and based on receiving the page-in request from the host OS, decrypting the contents of the physical memory page.

[0072] As discussed, the firmware injection component **207** facilitates firmware injection within the IVM architecture. As discussed in connection with phase **305** of an IVM guest lifecycle, the firmware injection component **207** copies a boot firmware image into the appropriate place in the IVM guest's GPA space, and the firmware injection component **207** (or the GPA integrity control component **205**) then changes the state of the GPA memory pages containing the boot firmware image to be accepted, thus allowing the IVM guest to access these boot firmware image memory pages. Thus, in some embodiments, method **400** further comprises, prior to receiving the acceptance request: populating one or more guest memory pages, within the GPA space, with guest boot firmware; and for each memory page in the one or more guest memory pages, setting a corresponding page acceptance indication to the accepted state.

[0073] As discussed, in embodiments, as part of unmapping an exclusive memory page from the IVM guest, MAC component **204** wipes the contents of the memory page (e.g., by zeroing the memory page, by writing a pattern of bits to the memory page, by writing random bits to the memory page, by deleting an encryption key associated with the memory page). Thus, in some embodiments, method **400** also comprises wiping contents of the physical memory page based on at least one of: the guest memory page being released to the host OS; or the isolated VM guest being shut down.

[0074] As discussed, the processor state isolation component **202** fully isolates an IVM guest's processor state from the host OS **116**, by preventing the host OS **116** from reading from or writing to an IVM guest's registers (including, e.g., the IVM guest's instruction and stack pointers). Thus, in some embodiments, method **400** also comprises isolating contents of one or more processor registers from the host OS.

[0075] Embodiments of the disclosure may comprise or utilize a special-purpose or general-purpose computer system that includes computer hardware (e.g., hardware **101**), such as, for example, one or more processors (e.g., processor **102**) and system memory (e.g., memory **103**), as discussed in greater detail below. Embodiments within the scope of the present disclosure also include physical and other computer-readable media for carrying or storing computer-executable instructions and/or data structures. Such computer-readable media can be any available media that can be accessed by a general-purpose or special-purpose computer system. Computer-readable media that store computer-executable instructions and/or data structures are computer storage media (e.g., storage media **104**). Computer-readable media that carry computer-executable instructions and/or data structures are transmission media. Thus, by way of example, embodiments of the disclosure can comprise at least two distinctly different kinds of computer-readable media: computer storage media and transmission media.

[0076] Computer storage media are physical storage media that store computer-executable instructions and/or data structures. Physical storage media include computer hardware, such as random access memory (RAM), read-only memory (ROM), electrically erasable programmable ROM (EEPROM), solid state drives (SSDs), flash memory, phase-change memory (PCM), optical disk storage, magnetic disk storage or other magnetic storage devices, or any other hardware storage device(s) which can be used to store program code in the form of computer-executable instructions or data structures, which can be accessed and executed by a general-purpose or special-purpose computer system to implement the disclosed functionality.

[0077] Transmission media can include a network and/or data links which can be used to carry program code in the form of computer-executable instructions or data structures, and which can be accessed by a general-purpose or special-purpose computer system. A “network” is defined as one or more data links that enable the transport of electronic data between computer systems and/or modules and/or other electronic devices. When information is transferred or provided over a network or another communications connection (either hardwired, wireless, or a combination of hardwired or wireless) to a computer system, the computer system may view the connection as transmission media. Combinations of the above should also be included within the scope of computer-readable media.

[0078] Further, upon reaching various computer system components, program code in the form of computer-executable instructions or data structures can be transferred automatically from transmission media to computer storage media (or vice versa). For example, computer-executable instructions or data structures received over a network or data link can be buffered in RAM within a network interface module, and then eventually transferred to computer system RAM and/or to less volatile computer storage media at a computer system. Thus, it should be understood that computer storage media can be included in computer system components that also (or even primarily) utilize transmission media.

[0079] Computer-executable instructions comprise, for example, instructions and data which, when executed at one or more processors, cause a general-purpose computer system, special-purpose computer system, or special-purpose processing device to perform a certain function or group of functions. Computer-executable instructions may be, for example, binaries, intermediate format instructions such as assembly language, or even source code.

[0080] It will be appreciated that the disclosed systems and methods may be practiced in network computing environments with many types of computer system configurations, including, personal computers, desktop computers, laptop computers, message processors, hand-held devices, multi-processor systems, microprocessor-based or programmable consumer electronics, network PCs, minicomputers, mainframe computers, mobile telephones, PDAs, tablets, pagers, routers, switches, and the like. Embodiments of the disclosure may also be practiced in distributed system environments where local and remote computer systems, which are linked (either by hardwired data links, wireless data links, or by a combination of hardwired and wireless data links) through a network, both perform tasks. As such, in a distributed system environment, a computer system may include a plurality of constituent computer systems. In a distributed system environment, program modules may be located in both local and remote memory storage devices.

[0081] It will also be appreciated that the embodiments of the disclosure may be practiced in a cloud computing environment. Cloud computing environments may be distributed, although this is not required. When distributed, cloud computing environments may be distributed internationally within an organization and/or have components possessed across multiple organizations. In this description and the following claims, “cloud computing” is defined as a model for enabling on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services). A cloud computing model can be composed of various characteristics, such as on-demand self-service, broad network access, resource pooling, rapid elasticity, measured service, and so forth. A cloud computing model may also come in the form of various service models such as, for example, Software as a Service (SaaS), Platform as a Service (PaaS), and Infrastructure as a Service (IaaS). The cloud computing model may also be deployed using different deployment models such as private cloud, community cloud, public cloud, hybrid cloud, and so forth.

[0082] Some embodiments, such as a cloud computing environment, may comprise a system that includes one or more hosts that are each capable of running one or more virtual machines. During operation, virtual machines emulate an operational computing system, supporting an OS and perhaps one or more other applications as well. In some embodiments, each host includes a

hypervisor that emulates virtual resources for the virtual machines using physical resources that are abstracted from view of the virtual machines. The hypervisor also provides proper isolation between the virtual machines. Thus, from the perspective of any given virtual machine, the hypervisor provides the illusion that the virtual machine is interfacing with a physical resource, even though the virtual machine only interfaces with the appearance (e.g., a virtual resource) of a physical resource. Examples of physical resources including processing capacity, memory, disk space, network bandwidth, media drives, and so forth.

[0083] Although the subject matter has been described in language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the described features or acts described above, or the order of the acts described above. Rather, the described features and acts are disclosed as example forms of implementing the claims.

[0084] The present disclosure may be embodied in other specific forms without departing from its essential characteristics. The described embodiments are to be considered in all respects only as illustrative and not restrictive. All changes which come within the meaning and range of equivalency of the claims are to be embraced within their scope.

[0085] When introducing elements in the appended claims, the articles “a,” “an,” “the,” and “said” are intended to mean there are one or more of the elements. The terms “comprising,” “including,” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements. Unless otherwise specified, the terms “set,” “superset,” and “subset” are intended to exclude an empty set, and thus “set” is defined as a non-empty set, “superset” is defined as a non-empty superset, and “subset” is defined as a non-empty subset. Unless otherwise specified, the term “subset” excludes the entirety of its superset (i.e., the superset contains at least one item not included in the subset). Unless otherwise specified, a “superset” can include at least one additional element, and a “subset” can exclude at least one element.

## Claims

1. A method, implemented at a computer system that includes a processor, for isolating resources of a virtual machine (VM) guest from a host operating system (OS), the method comprising: receiving an acceptance request from a guest partition corresponding to an isolated VM guest, the acceptance request identifying: a guest memory page that is mapped into a guest physical address (GPA) space of the guest partition, and a memory page visibility class, wherein the memory page visibility class is a shared read-only visibility class; determining that the physical memory page meets the memory page visibility class, including verifying, via a host OS second-level address translation table, that the host OS is granted read-only access to the physical memory page; and setting a page acceptance indication for the guest memory page from an unaccepted state to an accepted state based on a physical memory page mapped to the guest memory page meeting the memory page visibility class.

2. The method of claim 1, further comprising: detecting that the guest memory page has been mapped to a different physical memory page; and setting the page acceptance indication for the guest memory page to the unaccepted state.

3. The method of claim 1, further comprising, prior to receiving the acceptance request: detecting an access by the guest partition to an address covered by the guest memory page; and generating a page fault based on the page acceptance indication for the guest memory page being the unaccepted state.

4. The method of claim 1, further comprising, prior to receiving the acceptance request: populating a guest memory page, within the GPA space, with guest boot firmware; and setting a corresponding page acceptance indication to the accepted state.

5. The method of claim 1, further comprising verifying, via a host OS input/output memory



management unit table, that a direct memory access device is denied access to the physical memory page.

**6.** The method of claim 1, further comprising wiping contents of the physical memory page based on at least one of: the guest memory page being released to the host OS; or the isolated VM guest being shut down.

**7.** The method of claim 1, further comprising: based on receiving a page-out request from the host OS that identifies the physical memory page, recording integrity information for contents of the physical memory page; and based on receiving a page-in request from the host OS that includes an indication of the physical memory page, using the integrity information to verify the contents of the physical memory page.

**8.** The method of claim 7, further comprising: based on receiving the page-out request from the host OS, encrypting the contents of the physical memory page; and based on receiving the page-in request from the host OS, decrypting the contents of the physical memory page.

**9.** The method of claim 1, further comprising isolating contents of a processor register from the host OS.

**10.** A method, implemented at a computer system that includes a processor, for isolating resources of a virtual machine (VM) guest from a host operating system (OS), the method comprising: receiving an acceptance request from a guest partition corresponding to an isolated VM guest, the acceptance request identifying: a guest memory page that is mapped into a guest physical address (GPA) space of the guest partition, and a memory page visibility class, wherein the memory page visibility class is a shared read-write visibility class; determining that the physical memory page meets the memory page visibility class, including verifying, via a host OS second-level address translation table, that the host OS is granted read-write access to the physical memory page; and setting a page acceptance indication for the guest memory page from an unaccepted state to an accepted state based on a physical memory page mapped to the guest memory page meeting the memory page visibility class.

**11.** The method of claim 10, further comprising: detecting that the guest memory page has been mapped to a different physical memory page; and setting the page acceptance indication for the guest memory page to the unaccepted state.

**12.** The method of claim 10, further comprising, prior to receiving the acceptance request: detecting an access by the guest partition to an address covered by the guest memory page; and generating a page fault based on the page acceptance indication for the guest memory page being the unaccepted state.

**13.** The method of claim 10, further comprising, prior to receiving the acceptance request: populating a guest memory page, within the GPA space, with guest boot firmware; and setting a corresponding page acceptance indication to the accepted state.

**14.** The method of claim 10, further comprising verifying, via a host OS input/output memory management unit table, that a direct memory access device is denied access to the physical memory page.

**15.** The method of claim 10, further comprising wiping contents of the physical memory page based on at least one of: the guest memory page being released to the host OS; or the isolated VM guest being shut down.

**16.** The method of claim 10, further comprising: based on receiving a page-out request from the host OS that identifies the physical memory page, recording integrity information for contents of the physical memory page; and based on receiving a page-in request from the host OS that includes an indication of the physical memory page, using the integrity information to verify the contents of the physical memory page.

**17.** The method of claim 16, further comprising: based on receiving the page-out request from the host OS, encrypting the contents of the physical memory page; and based on receiving the page-in request from the host OS, decrypting the contents of the physical memory page.

**18.** The method of claim 10, further comprising isolating contents of a processor register from the host OS.

**19.** A computer system for isolating resources of a virtual machine (VM) guest from a host operating system (OS), comprising: a processor; and a computer storage media that stores computer-executable instructions that are executable by the processor to cause the computer system to at least: receive an acceptance request from a guest partition corresponding to an isolated VM guest, the acceptance request identifying: a guest memory page that is mapped into a guest physical address (GPA) space of the guest partition, and a memory page visibility class, wherein the memory page visibility class is a shared read-only visibility class or a shared read-write visibility class; determine that the physical memory page meets the memory page visibility class, including verifying, via a host OS second-level address translation table, one of: that the host OS is granted read-only access to the physical memory page when the memory page visibility class is the shared read-only visibility class; or that the host OS is granted read-write access to the physical memory page when the memory page visibility class is the shared read-write visibility class; and set a page acceptance indication for the guest memory page from an unaccepted state to an accepted state based on a physical memory page mapped to the guest memory page meeting the memory page visibility class.

**20.** The computer system of claim 19, wherein the computer-executable instructions are also executable by the processor to cause the computer system to: detect that the guest memory page has been mapped to a different physical memory page; and set the page acceptance indication for the guest memory page to the unaccepted state.

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